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Effects of different vitamin A supplies on performance and the risk of ketosis in transition cows

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ABSTRACT

This experiment investigated the effects of feeding low and high supplies of vitamin A (VA) during the transition period on plasma metabolites, prevalence of ketosis, and early milk production. In a randomized complete block design, 42 prefresh Holstein cows and 21 heifers were blocked by parity and calving date and assigned to 1 of 3 dietary treatments (n = 21 per treatment unless noted): CON, a transition diet with supplemental VA (75,000 IU/d) to meet the requirement; LVA, a transition diet with no supplemental VA; or HVA, a transition diet receiving supplemental VA (187,500 IU/d) 2.5 times greater than the requirement. Experimental periods were prepartum (−14 d prepartum), postpartum (1 to 30 d in milk), and carryover period (31 to 58 d in milk; common lactating diet with adequate VA was fed). Differences in dry matter intake in the pre- and postpartum periods and milk yield were not detected among treatment. Milk fat, protein, and lactose yields were similar among treatments and not affected by VA. Somatic cell count increased linearly with increasing VA. Body weight and body condition score decreased postpartum, but no VA effect was observed. Plasma retinol concentrations (n = 10 per treatment) decreased at d 2 postpartum and increased as lactation progressed, but the concentrations were unaffected by treatment. Plasma β-carotene (n = 10 per treatment) had a treatment by time interaction and its concentration decreased after parturition and remained low for 2 wk. Plasma fatty acids and β-hydroxybutyrate did not differ among treatments. Milk retinol concentration and yield (n = 10 per treatment) increased as VA supply increased. Segmented neutrophils (%) decreased, and lymphocytes (%) increased in blood with increasing VA supply. In conclusion, providing different supplies of VA did not affect production, mobilization of body fat, and risk of ketosis; however, excessive VA supply may have negatively affected the immune response, in part contributing to increased milk somatic cell counts during early lactation.

Key words: beta-carotene, prepartum, postpartum, carryover

INTRODUCTION

Nutrient demands increase dramatically at the onset of lactation, whereas feed intake lags behind (Grummer, 1995; Drackley, 1999). Consequently, fresh cows enter a state of negative energy balance (Grummer et al., 2004) where they mobilize body reserves to meet increased demands that cannot be met by feed intake (Tammingaa et al., 1997; Ingvartsen and Andersen, 2000). High rates of fat mobilization can lead to increased plasma nonesterified fatty acids (**NEFA**; Ingvartsen, 2006) and potential increases in blood ketone concentration. Accumulation of ketone bodies can result in ketosis during early lactation, which decreases milk production (Drackley, 1999; Herdt, 2000; Mulligan and Doherty, 2008), reproductive function (Walsh et al., 2007), and animal welfare and survivability (Mulligan and Doherty, 2008).

Ketosis primarily occurs during the first 2 to 7 wk postpartum, as negative energy balance is greatest in early lactation (Dohoo and Martin, 1984; Duffield et al., 1998). Ketosis can manifest as subclinical ketosis (**SCK**) and clinical ketosis. The prevalence of SCK can reach 60% of cows in a herd in the United States (Simensen et al., 1990; Duffield et al., 1998), and clinical ketosis can affect 2% to 15% of cows in a herd (Baird, 1982; Duffield, 2000). Ketotic cows are at higher risk of displaced abomasum, metritis, reduced fertility, and early removal from the herd (Ospina et al., 2010a; McArt et al., 2012a). Therefore, there is much interest in developing effective nutritional strategies to avoid high rates of body tissue mobilization postpartum without compromising milk production and farm profits.

Vitamin A (**VA**) is required for essential physiological functions including vision, cell growth, reproduction, and regulation and differentiation of epithelial

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and mesenchymal cells (NASEM, 2021). β-Carotene (**BC**), a precursor of VA, is also known for its contribution to VA in the body as well as its critical antioxidant capacity to eliminate radicals in the body (Sies et al., 1992), beneficial role in reproduction (Kawashima et al., 2009), and potential enhancement of immune system response (Bendich, 1991). In addition, VA supply can influence immune functions in lactating cows, but excess supply may be immune suppressive (NASEM, 2021). Other than those roles of VA, more attention has been paid recently to lipid-related gene regulation with VA at the transcriptional level (McGrane, 2007). Vitamin A is an important regulator of adipocyte differentiation as it can stimulate or inhibit adipocyte development depending on the VA status and stage of the adipogenic process in mice (Kuri-Harcuch, 1982; Schwarz et al., 1997; Ribot et al., 2001). In ruminants, VA-deficient diets (<30 IU/kg BW) increased the development of intramuscular adipose tissues compared with diets meeting VA requirements (Gorocica-Buenfil et al., 2007a,b). In contrast, excessive VA supplementation can potentially increase adipose tissue mobilization and oxidation (lipolysis) and adipocyte apoptosis (Kuri-Harcuch, 1982; Ribot et al., 2001; Felipe et al., 2003). Clearly, altering VA status can influence the adipocyte metabolism, therefore potentially affecting the degree of body fat mobilization of dairy cows during the transition period.

The objective of this experiment was to determine whether VA supplied at greater or lower amounts than the NRC (2001) requirement will alter blood plasma metabolites (NEFA and BHB), BCS, BW, production, and blood differential counts in transition dairy cows. We hypothesized that low VA supply would reduce body fat mobilization, which can lower prevalence of SCK. If fat mobilization is reduced with low VA supply, it would negatively affect milk production because of lack of energy. However, if the prevalence of SCK is reduced because of less fat mobilization, milk production could increase. In contrast, we hypothesized that excessive VA supply would accelerate body fat mobilization, increase SCK risk, and decrease production.

MATERIALS AND METHODS

Animals and Treatments

The experiment was conducted at Krauss Dairy Research Center, The Ohio State University (Wooster). All procedures involving the use of animals were approved by the Institutional Animal Care and Use Committee of The Ohio State University (IACUC:2019A00000104).

Forty-two prefresh Holstein cows and 21 heifers were used in a randomized complete block design and were blocked by parity (first vs. others) and expected calving date. The experiment consisted of 3 phases: prepartum, postpartum, and carryover period. Cows used in this experiment were dried off approximately 60 d before anticipated calving date and fed a common far-off diet with no supplemental VA until they entered the experiment. The lack of supplemental VA during the far-off period was intended to reduce hepatic retinol reserves. The far-off diet contained 24.8% corn silage, 27.7% alfalfa silage, 31% grass hay, 4.6% soybean meal, 9.7% ground corn grain, and a VA-free vitamin-mineral mix. Fourteen days before anticipated calving date, cows were moved into individual box stalls for individual feeding, where they were randomly assigned to 1 of 3 dietary treatments (prepartum period). Upon calving, cows were moved to individual tiestalls and fed the experimental postpartum diets (postpartum period). At the tiestalls, cows continued their assigned dietary treatment until 30 DIM. At 31 DIM, cows were moved to a common pen in a freestall barn and fed a common diet with adequate VA (carryover period). The carryover period lasted for 28 d (i.e., 31 to 58 DIM).

The basal diets during the prepartum and postpartum periods were formulated to meet or exceed all the nutrient requirements except VA according to NRC (2001; Tables 1 and 2). The 3 dietary treatments during the prepartum and postpartum periods were (1) control (**CON**), the basal diet to meet the requirement of VA (retinyl palmitate; 75,000 IU/d) according to NRC (2001); (2) low VA (**LVA**), the CON diet without supplemental VA; (3) high VA (**HVA**), the CON diet with supplemental VA (retinyl palmitate) 2.5 times greater than the requirement (187,500 IU/d), which was assumed to be excessive in supply of VA. Cows were fed once daily, and orts were recorded before feeding. During the prepartum period, cows were required to be on treatment at least 10 d before calving, except cows on LVA because they received a diet with no supplemental VA during the far-off dry period. The average days (mean ± SD) on treatment prepartum were LVA = 11 ± 3.8, CON = 14.4 ± 3.5, and HVA = 14.0 ± 3.8. Prepartum and postpartum cows were individually fed for an ad libitum intake with the target of 5% refusal. Cows were milked twice daily at 0400 and 1600 h.

Sampling and Measurements

Cows were weighed on 2 consecutive days before moving to box stalls (d 15 and 14 before anticipated calving), and on 2 and 3, 15 and 16, 29 and 30, and 57 and 58 DIM. Consecutive measures were averaged. The initial weight (average of d 15 and 14 prepartum) served as covariate. When cows calved, all calves were weighed within 24 h of birth and females were weighed

Table 1. Ingredients and chemical composition of prepartum diets

Item	Treatment[1]		
	LVA	CON	HVA
Ingredient, % DM			
Corn silage[2]	29.7	29.7	29.7
Rye silage[3]	25.6	25.6	25.6
Grass hay	29.5	29.5	29.5
Corn grain, ground	3.38	3.36	3.33
Soybean meal	6.08	6.08	6.08
Soybean hulls	3.20	3.20	3.20
Fat[4]	0.21	0.21	0.21
Mineral-vitamin mix[5]	2.35	2.35	2.35
Vitamin A	0.000	0.021	0.053
Chemical composition			
OM	93.0	93.1	92.6
CP	13.4	13.1	13.3
NDF	49.7	49.7	49.5
Fatty acids	2.34	2.31	2.27
Starch	12.6	12.3	12.5
Ca	0.45	0.50	0.47
P	0.31	0.31	0.31
Mg	0.25	0.24	0.24
K	2.05	2.06	2.05
Retinol, mg/kg DM	0.57	1.81	4.70
β-Carotene, mg/kg DM	11.3	11.2	11.2

[1]LVA = no supplemental vitamin A; CON = 75,000 IU/d vitamin A; HVA = 187,000 IU/d vitamin A.

[2]DM = 34.6%, CP = 6.5%, NDF = 37.4%, starch = 32.6% (DM basis).

[3]DM = 39.6%, CP = 11.3%, NDF = 62.0% (DM basis).

[4]True energy formula A (G. A. Wintzer and Son Company).

[5]The premix contains (as-is basis) 23% trace mineral salt, 14% magnesium sulfate, 7% magnesium oxide, 10% limestone, 7% selenium, 15% vitamin E IU/kg DM, 3% vitamin D 1,000 IU/kg DM, 20% biotin 100 mg/kg, 0.12% copper sulfate, and 0.34% zinc sulfate.

Table 2. Ingredients and chemical composition of postpartum diets

Item	Treatment[1]		
	LVA	CON	HVA
Ingredient, % DM			
Corn silage[2]	27.9	27.9	29.7
Alfalfa silage[3]	16.2	16.2	16.2
Grass hay	5.22	5.22	5.22
Corn grain, ground	21.3	21.3	21.2
Soybean meal	15.6	15.6	15.6
Soybean hulls	9.06	9.06	9.06
Fat[4]	1.22	1.22	1.22
Mineral-vitamin mix[5]	3.45	3.45	3.45
Vitamin A	0.00	0.01	0.04
Chemical composition			
OM	92.7	91.9	92.6
CP	16.5	16.4	16.5
NDF	32.1	31.1	31.0
Starch	24.4	23.7	23.7
Fatty acids	2.98	2.93	2.90
Ca	0.91	0.93	0.88
P	0.39	0.39	0.36
Mg	0.24	0.23	0.23
K	1.70	1.68	1.60
Retinol, mg/kg DM	0.47	2.50	5.37
β-Carotene, mg/kg DM	8.51	8.50	8.77

[1]LVA = no supplemental vitamin A; CON = 75,000 IU/d vitamin A; HVA = 187,000 IU/d vitamin A.

[2]DM = 34.6%, CP = 6.5%, NDF = 37.4%, starch = 32.6% (DM basis).

[3]DM = 49.2%, CP = 17.6%, NDF = 40.9% (DM basis).

[4]True energy formula A (G. A. Wintzer and Son Company).

[5]The premix contains (as-is basis) 29% trace mineral, 27% limestone, 3% selenium, 2% vitamin E IU/kg DM, 2% vitamin D 1,000 IU/kg DM, 16% dicalcium phosphate, 0.05% copper sulfate, 6% dynamite, 2% magnesium oxide, 13% biotin 100 mg/kg, and 0.1% zinc sulfate.

again at 42 d of age. No treatment was imposed on the calves. Body condition score (scale 1 to 5; Edmonson et al., 1989) was recorded for individual cows when moved to the box stalls (d 14 prepartum), at d 2 after parturition, at 30 DIM, and at the end of the carryover period (58 DIM). Three trained personnel assessed BCS and the scores were averaged.

Silages were sampled weekly and analyzed for DM at 100°C overnight to adjust their as-fed proportions in the diets when necessary. Weekly samples of forages and concentrates were collected and composited by month for chemical analysis (samples were frozen at −20°C until analysis). One refusal sample per cow during the prepartum period (composite sample from d 10 and 9 prepartum) and 2 during the postpartum (composite samples from 15 and 16 DIM and 29 and 30 DIM) were collected to analyze for DM (100°C overnight) to calculate daily DMI. Monthly composites of forages and concentrates were dried (55°C for 72 h) and ground through a 1-mm screen (Wiley Mill; Arthur A. Thomas Co.). Ground samples were then analyzed for DM (100°C overnight), ash (600°C overnight; AOAC International, 2000), NDF (Ankom Fiber Analyzer, Ankom Technology) with sodium sulfite and heat-stable α-amylase, long-chain fatty acids (Sukhija and Palmquist, 1988), N (Thermo Scientific Flash 2000 Element Analyzer), and minerals (infrared reflectance spectroscopy, Rock River Laboratory Inc.). Concentrates and corn silage samples were analyzed for starch (Holm et al., 1986). Additional forage and concentrate samples were collected monthly, vacuum sealed, and frozen at −20°C until analysis of retinol and BC using reverse-phase HPLC according to Weiss et al. (1995) with minor modifications. Briefly, dried and ground samples were mixed with 6% pyrogallic acid/ethanol solution. Then, 50% KOH was added, and the samples were incubated at 70°C for 30 min. Hexane was then added and samples were shaken for 5 min before being centrifuged for 10 min at 1,000 × g (4°C). The hexane layer was collected and evaporated in a dry block at 37°C under N_2 gas. Samples were reconstituted with ethanol and injected into the HPLC (PerkinElmer Series 2000 HPLC, Life and Analytical Sciences). The flow rate was 1.8 mL/min at ambient temperature. Retinol was detected using a fluorescence detector with emission at 344 nm and excitation at 472 nm (PerkinElmer 200 Series Fluorescence Detector) and BC was detected at 450 nm using a UV detector (PerkinElmer 785A UV/VIS Detector).

Weekly milk samples were collected for fat, lactose, and protein analysis (B2000 Infrared Analyzer, Bentley Instruments) by DHI Cooperative Inc. (Columbus, OH). Additional milk samples (a.m. and p.m.) were taken on 5 DIM, composited by morning and afternoon milk weights, and frozen at −20°C until analysis. An aliquot was processed according to the procedure by Indyk (1988) and analyzed for retinol and BC using reverse-phase HPLC as described earlier for feeds, except that milk samples were mixed with 1% pyrogallic acid/ethanol solution, 50% KOH, and placed in a water bath (70°C) for 7 min. Then, petroleum ether was added and centrifuged at 1,000 × g for 10 min (4°C). The organic layer was removed and dried under N_2 gas at 37°C. Samples were reconstituted with ethanol and injected into the HPLC. First milking colostrum was sampled within 12 h of calving. An aliquot was placed in aluminum weight boats and dried (100°C oven for 3 h) to determine solids content. Another aliquot was assayed for retinol and BC as just described. Colostrum samples were also analyzed for total N (Thermo Scientific Flash 2000 Element Analyzer) according to the procedure by Morris et al. (2019).

Blood samples were collected from the coccygeal tail vein on d 14 prepartum, and on 2, 5, 10, 15, and 30 DIM. Blood samples were collected 2 h postfeeding into sodium heparin vacutainer tubes (BD and Co.). Blood samples were centrifuged at 3,000 × g for 20 min (4°C), and plasma was collected and frozen at −20°C until analyzed. Plasma samples were assayed for BHB (Stanbio Laboratories, Boerne, TX), NEFA (Wako Diagnostics, Mountain View, CA), retinol, and BC (reverse-phase HPLC; Weiss et al., 1995). Threshold values for diagnosis of SCK were based on BHB ≥ 1,200 μmol/L (McArt et al., 2012b). Samples for retinol and BC analysis were mixed with ascorbic acid solution and hexane. Samples were centrifuged at 800 to 1,000 × g for 10 min (4°C). The organic top layer was removed and evaporated under N_2 gas. Samples were reconstituted with ethanol and injected into the HPLC. An additional blood sample was collected at 5 DIM into EDTA Vacutainer tubes (BD and Co.) and submitted to a commercial animal diagnostic laboratory (Petlabs Diagnostic Laboratories, Medina, OH) for a complete blood count and a differential leukocyte count.

Statistical Analysis

One cow receiving the HVA treatment died 3 d after calving for unknown reasons, and no data from the cow were included in the analysis. In total, 21 primiparous and 41 multiparous cows finished the trial, and all the dependent variables included data with n = 21 for CON and LVA and n = 20 for HVA except plasma retinol and BC, milk retinol and BC, and colostrum retinol and BC. For these variables, only 10 randomly selected blocks (multiparous cows) were used for analysis (i.e., n = 10 per treatment).

All statistical analysis was performed in SAS (version 9.4, SAS Institute Inc.) using the MIXED procedure. The statistical model for prepartum DMI and plasma concentrations for NEFA, BHB, retinol, BC, and BCS included the fixed effects of treatment, day, and treatment × day interaction, and the random effect of block. The model for postpartum DMI (weekly averaged), BW, milk yield (weekly averaged), and milk components data included the fixed effects of treatment, week, and treatment × week interaction, and the random effect of block. The model for colostrum weight, retinol and BC concentration and yield, and blood data for blood differential counts included the fixed effect of treatment and the random effect of block. Covariance structures for repeated measures (time effect) were selected according to the lowest Akaike information criterion. The spatial power covariance structure was used for repeated measures with unequal sampling time intervals. All degrees of freedom were calculated using the Kenward-Roger method. Polynomial contrasts were used to evaluate the effects of increasing VA (i.e., from LVA, CON, and HVA). Statistical significance was set as $P < 0.05$ and statistical trends were declared as $0.05 < P < 0.10$. All data are presented as least squares means ± largest standard error of mean.

RESULTS

DMI, BW, and BCS

Dry matter intake was not affected by VA in the pre- (average 10.6 kg/d) or postpartum period (18.8 kg/d), and no interaction of VA treatment by time was observed (Table 3). Body weight was not affected by VA, and no interaction of VA by time was found. All cows lost BW during the first 4 wk of lactation (LVA = −3.7 kg/d, CON = −4.1 kg/d, and HVA = −3.5 kg/d; linear $P = 0.66$; data not shown). Average BCS was not affected by VA, and no VA by day interaction for BCS was observed. Collectively, all cows lost BCS from d 14 before calving until d 58 of lactation and decreased BCS by 0.48, 0.42, and 0.33 units for CON, LVA, and HVA, respectively (data not shown). Calf weights at birth and 42 d were not affected by treatments (data not shown).

Table 3. Effects of feeding diets with different levels of vitamin A supplementation to transition cows (14 d prepartum to 30 d postpartum) on DMI prepartum and postpartum and on BCS and BW of transition cows

	Treatment[1]				P-value[2]	
Item	LVA	CON	HVA	SEM	Linear	Quadratic
Prepartum DMI, kg/d	10.7	10.6	10.6	0.370	0.75	0.84
Postpartum DMI, kg/d	19.1	19.2	18.2	0.780	0.22	0.50
BCS[3]	3.21	3.29	3.29	0.046	0.20	0.28
BW,[4] kg	662	662	665	5.810	0.66	0.89

[1]LVA = no supplemental vitamin A (n = 21); CON = 75,000 IU/d vitamin A (n = 21); HVA = 187,000 IU/d vitamin A (n = 20).

[2]DMI, BCS, and BW were affected by time ($P < 0.01$): day for prepartum DMI and BCS, and week for postpartum DMI and BW.

[3]Scored on 2 and 30 DIM.

[4]Measured on −15 and −14 d before calving, and 2 and 3, 15 and 16, and 29 and 30 d after calving.

Milk Yield and Composition

The VA supply had no effect on milk yield during the postpartum period (Table 4). Milk fat, protein, and lactose yields did not differ among treatments during the postpartum period. However, VA supplementation had a linear effect on milk fat percentage ($P = 0.02$), where increasing VA increased milk fat content during the postpartum period. Milk lactose percentage tended to decrease ($P = 0.07$) linearly as VA supplementation increased. Feed efficiency and ECM were not affected by dietary treatments. Milk urea N concentration did not differ among treatments. However, SCC ($\log_{10}$ transformed) increased ($P < 0.01$) as VA increased. None of the variables had an interaction of VA by time. During the carryover period from 31 to 58 DIM (Table 5), milk yield and milk composition were not affected by VA except that milk protein content showed a quadratic trend ($P = 0.09$) where cows on CON had the lowest content (2.86%) compared with LVA (2.90%) and HVA (2.94%). As similar in the postpartum period, SCC (log-transformed) increased linearly ($P < 0.01$) as VA supplementation increased during the carryover period. An interaction of VA by time was not observed for all variables in the carryover period except that an interaction trend ($P = 0.09$) for milk fat yield was observed.

Plasma Metabolite Concentrations

Plasma concentrations of retinol were not affected by VA (n = 10 per treatment; Table 6). There was no VA by time interaction. Retinol plasma concentrations decreased as cows approached calving and gradually increased as lactation progressed (d 5 to 30 of lactation) for all treatments (Figure 1). Plasma concentration of BC (n = 10 per treatment) decreased after parturition

Table 4. Effects of feeding diets with different levels of vitamin A supplementation to transition cows (14 d prepartum to 30 d postpartum) on milk production and milk components of transition cows during the postpartum period (1 to 30 DIM)

	Treatment[1]				P-value[2]	
Item	LVA	CON	HVA	SEM	Linear	Quadratic
Milk yield, kg/d	37.4	38.0	35.8	1.770	0.24	0.35
Milk yield/DMI, kg/kg	1.96	2.00	1.96	0.053	0.90	0.57
ECM,[3] kg/d	41.6	43.5	42.4	1.680	0.67	0.22
ECM/DMI, kg/kg	2.22	2.33	2.38	0.074	0.14	0.64
Milk fat, kg/d	1.61	1.74	1.74	0.073	0.19	0.26
Milk protein, kg/d	1.18	1.18	1.14	0.061	0.44	0.71
Milk lactose, kg/d	1.82	1.83	1.72	0.092	0.20	0.48
Milk fat, %	4.49	4.77	5.12	0.206	0.02	0.90
Milk protein, %	3.18	3.13	3.21	0.047	0.56	0.30
Milk lactose, %	4.85	4.82	4.78	0.027	0.07	0.91
MUN, mg/dL	11.0	10.1	10.8	0.560	0.90	0.18
SCC, $\log_{10}$	4.76	4.75	5.26	0.113	<0.01	0.11

[1]LVA = no supplemental vitamin A (n = 21); CON = 75,000 IU/d vitamin A (n = 21); HVA = 187,000 IU/d vitamin A (n = 20).

[2]All variables were affected by time ($P < 0.01$) except MUN (mg/dL).

[3]ECM = (0.327 × kg milk) + (12.95 × kg milk fat) + (7.2 × kg milk protein).

Table 5. Effects of feeding diets with different levels of vitamin A supplementation to transition cows (14 d prepartum to 30 d postpartum) on milk production and milk components of lactating cows during the carryover period (31 to 58 DIM)

	Treatment[1]				*P*-value[2]	
Item	LVA	CON	HVA	SEM	Linear	Quadratic
Milk yield, kg/d	43.7	45.3	41.9	2.070	0.27	0.17
ECM, kg/d	46.2	47.3	44.6	1.880	0.31	0.28
Milk fat,[3] kg/d	1.75	1.79	1.70	0.072	0.47	0.46
Milk protein, kg/d	1.28	1.29	1.24	0.061	0.40	0.50
Milk lactose, kg/d	2.17	2.24	2.07	0.102	0.23	0.20
Milk fat, %	4.08	4.04	4.18	0.152	0.55	0.66
Milk protein, %	2.90	2.86	2.94	0.036	0.25	0.09
Milk lactose, %	4.96	4.94	4.92	0.021	0.13	0.94
MUN, mg/dL	10.8	10.5	10.5	0.610	0.69	0.75
SCC, $\log_{10}$	4.35	4.38	4.89	0.126	<0.01	0.20

[1]LVA = no supplemental vitamin A (n = 21); CON = 75,000 IU/d vitamin A (n = 21); HVA = 187,000 IU/d vitamin A (n = 20).

[2]All variables were affected by time ($P < 0.01$) except MUN (mg/dL).

[3]Treatment × week interaction trend ($P = 0.09$).

and was relatively constant until 15 DIM for all treatments (Figure 2). After that, BC concentration gradually increased as lactation progressed until 30 DIM. Plasma BC had a VA by time interaction ($P = 0.03$), which occurred because BC concentration became greater with decreasing VA supply as lactation progressed, which was not clearly observed until 10 DIM.

Concentrations of NEFA in plasma were not affected by VA supplies (Table 6). Concentrations of NEFA increased at d 2 after parturition for all 3 treatments and remained elevated until d 5 postpartum for CON and LVA treatment (data not shown). Afterward, NEFA concentrations gradually decreased for all treatments by 30 d postpartum. Plasma BHB concentrations were not affected by VA (Table 6). Generally, concentrations gradually increased after parturition by 30 DIM (data not shown). No interaction of VA by time was observed for NEFA and BHB.

Milk and Colostrum VA

The weight of colostrum was not affected by VA treatment (Table 7). No VA effect was observed on concentrations of retinol in colostrum (on average 1.86 μg/mL; n = 10 per treatment). Colostrum retinol yield was not different among treatments either. A quadratic trend ($P = 0.06$) was observed for BC concentrations in colostrum (n = 10 per treatment), as it was greater for HVA (1.33 μg/mL) compared with LVA (1.13 μg/mL) and CON (1.09 μg/mL); however, the yield of colostrum BC did not differ. Retinol concentration in milk (5 DIM; n = 10 per treatment) increased linearly ($P = 0.04$) as VA supply increased (Table 7). Similarly, milk retinol yield tended to increase ($P = 0.08$) as VA supply increased. Concentration and yield of milk BC (n = 10 per treatment) were not affected by VA treatment (Table 7).

Table 6. Effects of feeding diets with different levels of vitamin A supplementation to transition cows (14 d prepartum to 30 d postpartum) on plasma retinol, β-carotene, nonesterified fatty acid (NEFA), and BHB concentrations of transition cows during the postpartum period

	Treatment[2]				*P*-value[3]	
Item[1]	LVA	CON	HVA	SEM	Linear	Quadratic
Retinol, μg/mL	0.20	0.20	0.20	0.013	0.89	0.87
β-Carotene,[4] μg/mL	1.22	1.24	0.96	0.131	0.14	0.45
NEFA, μEq/L	331	324	314	24.800	0.60	1.00
BHB, μmol/L	1,004	1,027	894	107.000	0.35	0.54

[1]Retinol and β-carotene, n = 10 for LVA, CON, and HVA; NEFA and BHB, n = 21 for LVA and CON, and n = 20 for HVA. Plasma samples were collected on 2, 5, 10, 15, and 30 DIM.

[2]LVA = no supplemental vitamin A; CON = 75,000 IU/d vitamin A; HVA = 187,000 IU/d vitamin A.

[3]All variables were affected by time ($P < 0.05$).

[4]Treatment × day interaction ($P = 0.03$).

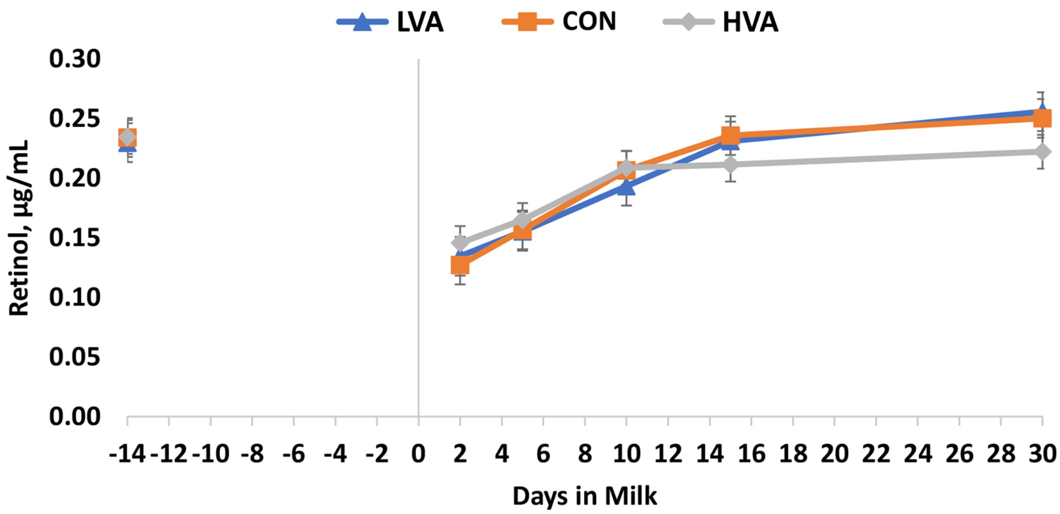

Figure 1. Effects of feeding diets with different levels of vitamin A supplementation to transition cows (14 d prepartum to 30 d postpartum) on concentration of retinol in plasma of transition cows at d 14 before calving and at 2, 5, 10, 15, and 30 DIM. Vertical line represents parturition (0 DIM). LVA = no supplemental vitamin A (n = 10); CON = 75,000 IU/d vitamin A (n = 10); HVA = 187,000 IU/d vitamin A (n = 10). Linear effect of VA, $P = 0.89$; quadratic effect of VA, $P = 0.87$; VA × d, $P = 0.59$. Error bars indicate SEM.

Blood Differential Count

Cows fed increasing VA supply had a decreased ($P = 0.01$) segmented neutrophil percentage in blood 5 d after parturition (Table 8). Complementarily, lymphocyte percentage increased with increased supplementation of VA ($P = 0.01$). No other variables from the blood differential count were unaffected by VA treatment.

DISCUSSION

The aim of this study was to determine whether the degree of fat mobilization and risk of ketosis during early lactation is affected by supplementing different amounts of VA, and whether this response affects milk production of postpartum cows. We intended to decrease VA stores in cows on LVA during the experiment to test the hypothesis, for which we provided a far-off

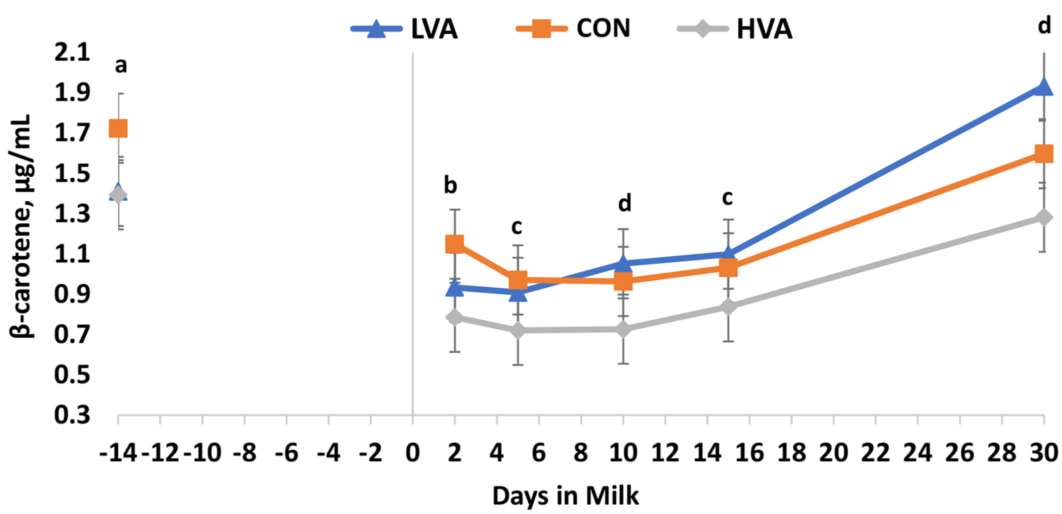

Figure 2. Effects of feeding diets with different levels of vitamin A supplementation to transition cows (14 d prepartum to 30 d postpartum) on concentration of β-carotene in blood plasma of transition cows at d 14 before calving and at 2, 5, 10, 15, and 30 DIM. Vertical line represents parturition (0 DIM). LVA = no supplemental vitamin A (n = 10); CON = 75,000 IU/d vitamin A (n = 10); HVA = 187,000 IU/d vitamin A (n = 10). Linear effect of VA, $P = 0.14$; quadratic effect of VA, $P = 0.45$; VA × d, $P = 0.02$. a = quadratic trend ($P < 0.10$); b = quadratic effect ($P < 0.05$); c = linear trend ($P < 0.09$); d = linear effect ($P < 0.05$). Error bars indicate SEM.

Table 7. Effects of feeding diets with different levels of vitamin A supplementation to transition cows (14 d prepartum to 30 d postpartum) on colostrum and milk retinol and β-carotene concentration and yield of transition cows during the postpartum period

	Treatment[1]				P-value	
Item	LVA	CON	HVA	SEM	Linear	Quadratic
Colostrum						
Weight, kg/d	7.20	7.31	5.25	1.187	0.22	0.54
Concentration, μg/mL						
Retinol	1.89	1.61	2.08	0.262	0.53	0.28
β-Carotene	1.13	1.09	1.33	0.165	0.38	0.06
Yield, mg/d						
Retinol	13.5	11.5	10.7	2.680	0.48	0.79
β-Carotene	8.33	7.83	6.00	1.529	0.22	0.79
Milk[2]						
Concentration, μg/mL						
Retinol	0.30	0.32	0.37	0.026	0.04	0.64
β-Carotene	0.13	0.12	0.12	0.012	0.54	0.99
Yield, mg/d						
Retinol	10.5	11.3	12.8	0.940	0.08	0.93
β-Carotene	4.35	4.36	4.01	0.408	0.48	0.74

[1]LVA = no supplemental vitamin A (n = 10); CON = 75,000 IU/d vitamin A (n = 10); HVA = 187,000 IU/d vitamin A (n = 10).

[2]Samples collected at 5 DIM.

diet with no supplemental VA to all cows during the entire far-off period (approximately 45 d) before cows entered the experiment. This was necessary because liver and adipose tissues store large amounts of VA that are released and used in the body when dietary VA is insufficient (Blaner et al., 2016). However, according to previous studies with beef cattle, 45 d is likely not sufficient to cause VA deficiency in the body (see more details later in this discussion). Nevertheless, as dairy cows require more VA at parturition compared with beef cattle because they produce colostrum and milk, we anticipated a decrease in VA stores large enough to alter lipid metabolism for cows on LAV during the postpartum period.

Table 8. Effects of feeding diets with different levels of vitamin A supplementation to transition cows (14 d prepartum to 30 d postpartum) on the blood differential count test of transition cows at d 5 postpartum

	Treatment[1]				P-value	
Item	LVA	CON	HVA	SEM	Linear	Quadratic
Absolute white cell count, 10^3/μL	9.01	10.11	10.32	0.856	0.28	0.55
Absolute segmented neutrophil, 10^3/μL	4.51	4.34	3.94	0.471	0.47	0.53
Absolute banded neutrophil, 10^3/μL	0.00	0.01	0.02	0.010	0.25	0.86
Absolute monocyte, 10^3/μL	0.62	0.73	0.64	0.091	0.96	0.34
Absolute eosinophil, 10^3/μL	0.17	0.13	0.16	0.046	0.96	0.43
Red blood cell count, 10^6/μL	6.15	5.79	6.04	0.158	0.72	0.07
Hemoglobin, g/dL	10.8	10.6	10.7	0.200	0.84	0.39
Hematocrit, %	31.1	30.5	31.4	0.610	0.64	0.29
Mean corpuscular volume, fL	50.9	52.9	52.3	0.890	0.34	0.19
Mean corpuscular hemoglobin, pg	17.7	18.5	17.9	0.340	0.91	0.08
Mean corpuscular hemoglobin concentration, g/dL	34.8	34.9	34.2	0.400	0.25	0.45
Platelet count, 10^3/μL	250	240	227	36.500	0.62	0.94
Segmented neutrophil, %	49.5	47.0	38.7	3.760	0.01	0.62
Banded neutrophil, %	0.09	0.20	0.27	0.189	0.43	0.84
Lymphocyte, %	41.2	43.6	52.7	4.060	0.01	0.56
Monocyte, %	6.91	7.46	6.60	0.828	0.71	0.47
Eosinophil, %	1.94	1.59	1.66	0.471	0.69	0.66
Basophil, %	0.20	0.15	0.28	0.147	0.66	0.64

[1]LVA = no supplemental vitamin A (n = 21); CON = 75,000 IU/d vitamin A (n = 21); HVA = 187,000 IU/d vitamin A (n = 20).

VA and BC Plasma Concentrations

Concentrations of plasma retinol decrease during late gestation and reach a nadir shortly after parturition before increasing to the baseline as lactation progresses (Oldham et al., 1991; Michal et al., 1994; Goff et al., 2002). This is mainly related to increased VA uptake by the mammary gland to secrete into colostrum and milk production (Goff and Stabel, 1990). In our study, plasma retinol concentrations decreased around parturition, reaching a nadir at d 2 postpartum, before rising to initial values as lactation progressed until 30 DIM. The lack of difference in plasma retinol concentrations among treatments is likely due to the tight regulation of plasma retinol concentrations, which are controlled by the regulation of retinol-binding protein in the liver (Thurnham and Northrop, 1999). Hepatic retinol stores need to be depleted for blood retinol concentrations to be suppressed. Therefore, unless hepatic retinol stores are exhausted, blood VA concentrations can be relatively maintained to support body functions, although infections can alter this homeostatic control of plasma VA (Johnston and Chew, 1984; Akar and Gazioglu, 2006; Strickland et al., 2021). Therefore, the lack of difference in plasma retinol concentration in the current study suggests that cows on LVA did not achieve a deficient body VA status. The results are in line with results of serum VA concentrations in Agustinho et al. (2021). In that study, a diet with no supplemental VA versus a diet adequate in VA was compared in transitioning cows (−35 d to 21 d relative to calving) and no difference in serum VA was found. Oka et al. (1998) conducted a study in feedlot beef cattle and observed a decline in plasma retinol concentration from 0.4 μg/mL to 0.07 μg/mL after receiving a VA-free diet for 15 mo. In that study, animals fed the VA-free diet had visibly greater intramuscular fat compared with animals treated with a VA injection every 2 mo (303 mg of retinol; equivalent 35,000 IU/kg of BW), which maintained plasma VA concentrations above 0.3 μg/mL. Similarly, Gorocica-Buenfil et al. (2007c) reported a decrease in plasma retinol concentration in Holstein steers fed a restricted VA diet (no supplemental VA) long term (243 d) or short term (131 d) compared with control (2,200 IU/kg of DM for 243 d). Plasma retinol declined to 0.212 and 0.252 μg/mL for long- and short-term restriction, respectively, compared with 0.369 μg/mL for control treatment. Follow-up studies (Gorocica-Buenfil et al. 2007a,b,c, 2008) have concluded that feeding a low-VA diet (no supplemental VA) for more than 168 d, but not for less than 145 d, can decrease serum and hepatic retinol concentrations in beef cattle. Therefore, feeding a diet with no supplemental VA for only 45 d may have decreased VA stores in LVA cows but did not result in deficiency of VA in the body, although dairy cows required more VA than beef cattle when lactation began in the current study. Another possibility is that cows on LVA did not reach the depletion of VA because of potential supply of VA from BC (see more discussion later).

Previous studies reported decreases in plasma BC at parturition until 7 d after calving in dairy cows (Daniel et al., 1991; Chawla and Kaur, 2004), and plasma BC started increasing at 2 wk after calving (Goff et al., 2002). The decrease in plasma BC was partly attributed to the increased output of BC in colostrum and milk (Goff et al., 2002). However, BC has other roles in reproduction, immunity, and antioxidant capacities (Nozière et al., 2006). After parturition, BC concentrations could remain low because of increased oxidation of BC by the oxidative stress imposed by calving (Goff et al., 2002; Chew and Park, 2004). Furthermore, the decrease in plasma BC concentrations and consequent rise in plasma retinol after parturition may indicate that BC is converted to retinol (Nozière et al., 2006; Bohn et al., 2019). In the current study, therefore, the decrease in plasma BC concentrations for 2 wk after calving and consequent rise in plasma retinol after parturition may also suggest that BC was used to supply retinol. In part, this may explain the lack of VA deficiency for LVA and the lack of difference in plasma VA concentrations among treatments. We observed a treatment by time interaction for BC where plasma BC concentration increased in a linear fashion as VA supply decreased. The linear increases became more obvious as lactation progressed (i.e., from d 10 to 30 in milk) and when plasma BC at −14 d was used as a covariate where BC concentration tended to be linear ($P = 0.06$) and was linear ($P < 0.05$) from 5 to 30 DIM (data not shown). The observed increase in plasma BC as VA supply decreased may indicate either inefficient utilization of BC or additional supply of BC to cells to compensate for VA shortage (Tang, 2010). More studies are needed to understand changes in blood BC concentration in relation to VA supply. If the latter was the case in the current study, however, it is unclear whether the increased plasma BC concentration as VA supply decreased occurred from enhanced BC absorption from the feed or whether the body spared BC by reducing its own functions in the body (e.g., antioxidant activity, reproductive functions, fertility; Hurley and Doane, 1989; van den Berg et al., 2000).

Plasma Metabolite Concentrations

Circulating NEFA in blood reflects the degree of body fat mobilization. In the present study, plasma NEFA concentrations were within the normal range in both the pre- (200 to 300 μEq/L) and postpartum periods (<700 μEq/L; Drackley et al., 2001). Plasma concentrations of NEFA did not differ among VA treatments, which is also contrary to our hypothesis. This also supports the failure of inducing a deficient VA status or decreasing VA stores enough to alter lipid metabolism in cows on LVA, probably because the length of VA shortage was insufficient or the utilization of BC to support VA was enhanced, as previously discussed.

Plasma concentrations of BHB are another biological indicator of ketosis (Duffield et al., 1998). McArt et al. (2012b) observed that a higher incidence of SCK (BHB ≥ 1,200 μmol/L) exists in the first week of lactation (3 to 16 DIM), with peak incidence occurring at 5 DIM. The study also reported that cows that developed SCK within their first week postpartum were more likely to suffer from adverse health events and decreased milk production. In the present study, the SCK incidence (BHB ≥ 1,200 μmol/L) was 21% across all treatments (data not shown) in the first 15 DIM, with peak occurrence at 5 DIM, and the incidence was similar among treatments. The prevalence reported in this study is lower than previously reported for SCK incidences of 40% to 60% in Canadian and US dairy farms (Duffield et al., 1998; McArt et al., 2012a,b, 2013). Similar to NEFA concentrations, BHB concentrations observed were contrary to our hypothesis and confirm that different VA supplies did not influence body fat mobilization during early lactation.

BCS Changes

The extent of body tissue mobilization has been related to several factors such as productivity (Erdman and Andrew, 1989) and BCS at calving (Hart et al., 1979). In the present study, cows entered the prepartum period (14 d before anticipated calving date) with an average BCS of 3.25. The BCS range from 3.0 to 3.5 is considered ideal for optimal milk production and reproductive success and to prevent cows from suffering from metabolic diseases and health problems around parturition (Markusfeld et al., 1997; Gillund et al., 2001; Roche et al., 2004). In our study, cows collectively lost less than half a point (0.40) unit of BCS across treatments, which agrees with Roche et al. (2009) and Buckley et al. (2003). Neither BCS nor BCS loss differed among treatments, suggesting that different VA supplies in the current study did not affect body fat mobilization.

Milk Production and Performance

The effect of VA on lactation performance has been the focus of various studies with VA in dairy cows. Michal et al. (1994) reported no difference in milk production when transition cows were fed no supplemental VA, supplemented VA (120,000 IU/d), 300 mg/d of BC, or 600 mg/d of BC starting at 4 wk prepartum and continuing until 4 wk postpartum. Jin et al. (2014) provided midlactation cows with the requirement for supplemental VA (110 IU/kg of BW; NRC, 2001) or a high dose of VA supplementation (220 IU/kg BW) for 60 d and observed no VA effect on milk production except for an increase in milk VA content when supplemented with the high dose. Our study agrees with Jin et al. (2014) where increasing VA supplementation linearly increased VA concentration and yield in milk. Similarly, milk yield and composition were not affected by VA supplies in our study. This is further supported by the lack of significant difference observed in DMI, BCS, and blood metabolites among treatments. Despite no difference in milk composition and yield, there were increases in the concentration of fat with increasing VA supplementation in the postpartum period (30 DIM) and carryover period (31 to 58 DIM). However, the lack of difference in fat yield suggests that the difference in concentrations occurred because of dilution where milk yield for HVA was numerically lower than LVA and CON. The lack of difference in production measures postpartum led to no effects of VA supply on production during the carryover period.

Milk SCC and Blood Differential Count

A linear increase in SCC during the postpartum period (1 to 30 DIM) and carryover period (31 to 58 DIM) in cows receiving increasing VA supply was observed. Generally, it is well appreciated that VA affects the immune system and its response to infections (Huang et al., 2018), where adequate supply of VA can be beneficial but excess supply of VA might be immune suppressive according to studies with nonruminants and humans (NASEM, 2021). In terms of responses of SCC to VA supply in lactating cows, providing VA greater than the requirement generally did not increase SCC in previous studies. Johnston and Chew (1984) reported that elevated feeding of BC from 3 wk before dry-off through the first 10 wk of the next lactation lowered SCC during early lactation compared with cows not supplemented with VA. Puvogel et al. (2005) supplemented cows with VA from dry-off until 100 DIM with 880 IU/kg BW (8 times the requirement) and observed no difference in SCC compared with cows receiving no supplemental VA. Jin et al. (2014) reported that feeding VA at 2 times greater than the requirement decreased SCC in midlactation cows compared with cows fed at the VA requirement. The result of the study is contrary to that of the current study, where HVA increased SCC.

In addition, Agustinho et al. (2021) provided to transition cows (−35 d before calving to 21 d after calving) a diet without supplemental VA and found increases in SCC for no supplemental VA compared with a diet adequate in VA, which was not observed in the current study. Although studies on SCC with VA supply are limited, results have been variable, and more studies are needed to understand the relationship.

When the SCC data are examined with the blood cell counts, there is a clear need to better understand how VA supplementation affects the immune system of transitioning cows. The moderate, but still unclear, increase in SCC with cows on HVA was associated with a greater abundance of lymphocytes and a decreased abundance of neutrophils relative to total leukocytes. Together, these data may indicate that different VA supplies altered the immune status, leaving the cows either more or less resistant to infectious disease. Although no clinical diseases were noted, we did not explicitly seek out diagnoses for the multitude of subclinical diseases that may have affected the results. In addition, it is worth noting that the time measure of blood parameters during the postpartum period in the current study needs to be interpreted with caution.

CONCLUSIONS

Restricting VA from the far-off dry period until early lactation (30 DIM) probably did not create a VA deficiency in cows or did not decrease VA stores sufficiently in the body to alter lipid metabolism (i.e., fat mobilization). The lack of VA deficiency may have occurred due to insufficient time of VA restriction or enhanced utilization of BC to support VA. Therefore, VA had no effect on changes in BW and BCS, which is also supported by the lack of difference in plasma NEFA and BHB concentrations. The increased SCC in milk and decrease in neutrophil percentage in blood observed for HVA warrants additional investigation, as it may be indicative of increased susceptibility to infections when VA is provided in excess during the transition period. Further studies should focus on the required time for low VA supply to deplete VA stores in dairy cows as well as the effects of oversupplementing VA on the immune system response during the transition period.

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